张镇晖

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性别: 男 | 籍贯: 福建福州 | 民族: 汉族 | 政治面貌: 共青团员

教育经历

西安电子科技大学 2019.09 - 2022.07
航天工程 硕士 空间科学与技术学院 西安
平均成绩：90.03/100

中国民航大学 2015.09 - 2019.08
通信工程 本科 电子信息与自动化学院 天津
GPA：3.46/4.0

技能/证书及其他

外语：CET-4 PETS-3；计算机：C 语言二级 网络技术三级；普通话：二级乙等；制图员：四级

荣誉奖项

全国大学生工程训练综合能力竞赛全国一等奖 2017.05
全国大学生电子设计竞赛天津市一等奖/三等奖 2017.12/2016.12
"北斗杯"青少年科技大赛全国二等奖/优秀奖 2017.05/2016.05
霍尼韦尔“星”计划最佳创意奖/最佳动手能力奖 2017.07/2016.07
NXP 全国大学生智能车大赛华北赛区二等奖 2017.08
天津市大学生工程训练综合能力竞赛天津市二等奖 2017.03
天津市机器人大赛天津市一等奖 2016.10
华北五省机器人大赛二等奖/三等奖 2017.11/2016.11

项目经历

科学电控主控设备 2019.12 - 2021.07
中国科学院上海光学精密机械研究所 上海 硬件设计负责人

- **项目介绍：**该设备将用于中国空间站实验舱Ⅱ"梦天"中的超冷原子物理科学实验柜，利用微重力环境优势，创造新的极端条件，获得地面无法达到的 10-12 K（即 pK）量级的超低温，建设具有超低温、大尺度、高质量，适合长时间精密测量的玻色与费米量子简并工作物质的开放实验系统。该控制器是实验系统中的主要控制模块，主要实现对激光子系统和物理控制单元控制指令的发送，以及对实验数据和图像的采集与处理等功能。多个板卡以 VPX 结构互联，使用了多块 Xilinx 公司的 ZYNQ Ultrascale 系列 FPGA 作为处理控制设备，通过网络传送激光子系统和物理控制单元所需的配置指令和控制指令，并通过 USB-CCD 相机采集实验图像，转换为 HDMI 信号输出到显示屏。还通过高速 ADC、DAC、DDS 器件的协调配合，通过 AOM 器件监控激光光强与频率，再通过调控 DAC 与 DDS 器件的输出，实现对激光器的光强与频率的调整稳定。

- **主要工作：**作为项目硬件负责人，完成所有板卡的原理图绘制工作及部分 PCB 的布局布线，包括 FPGA 供电电路，网口 PHY 电路，USB 电路；完成了 ZYNQ 器件的硬件逻辑开发工作，使用 Verilog 语言并基于 AXI 总线编写自定义 IP，加以 AXI-DMA 互联 IP，实现了多路 USB、网口，hdmi 显示，adc、

dac、ttl、uart、spi 多种高低速接口的高精度并行工作。并且，还根据航天项目的特殊要求，对电路设计、代码编写进行了工程化处理与验证工作。

超高速数据采集与持续存储系统的研究与实现 2019.06 - 2020.05

中国空间技术研究院西安分院(504 所) 西安 FPGA 硬件开发

- **项目介绍：**该系统以 2 块 Xilinx 公司 Virtex7 系列 FPGA 为平台，利用 2 块采样率可达 5Gsps、数据位宽 8bit 的高速 ADC 进行模拟信号采样，实时将 80Gbps 的采样数据通过 PCIe 传输到计算机内存供后端处理。

- **主要工作：**作为项目组成员，完成了硬件逻辑开发。使用 2 套 VC709 开发板和 4 块 DSP 公司 5Gsps ADC 的 FMC 子板进行开发和调试。对单套 FPGA 开发板：使用 IDELAYE 和 ISERDERS 进行高速数据的同步，设计时钟边沿对齐算法消除亚稳态；针对数据不齐设计多通道数据自动对齐算法；采用数据拼接的思想对 FPGA 接收到的高速 LVDS 数据进行重组；消除实际中各个 ADC 的偏置误差、增益误差和时钟相位误差等保证等间隔采样；使用 MIG IP 核实现高速率 DDR3 芯片的控制设计，满足大容量数据的高速率存储和读取要求；消除跨时钟域高速并行数据传输的异步问题；同组内成员配合，利用 Xilinx 的 DMA for PCIe IP 核进行 PCIe Gen3 高速数据传输开发，实现单板 40Gbps 稳定数据传输及数据落盘。利用并行交替采样原理输入给两板不同相位的时钟，在计算机中对两个板卡采集后的数据进行同步，实现整个系统 80Gbps 数据采集。

A320-IPT 模拟机 2016.09 - 2019.06

中国民航大学 天津 项目负责人

- **主要工作：**大学期间参与 A320-IPT 模拟机的研制开发工作，并为国家级项目“基于 FSUIPC 的简易飞机模拟机平台的搭建” 的项目负责人，并发表论文一篇。 截至目前，已完成马鞍架的设计加工，A320 油门仿制，并结合 FSUIPC 协议实现与 P3D 飞行模拟软件的对接。

专业技能

- 熟练使用 Vivado 软件与 Verilog 硬件描述语言进行 FPGA 器件的开发工作，掌握 Xilinx 公司相关 FPGA、ZYNQ 器件的开发流程，具备较为丰富的 FPGA 器件设计开发经历；
- 熟练掌握多类器件的原理图设计，能够独立进行 Multisim 电路仿真与 AD 电路板的设计与绘制，具备较为丰富的电路板设计开发经历；
- 熟悉 Linux 操作系统的基础操作与 Petalinux 系统的开发设计流程，掌握 51 系列、ardrino、STM32 系列单片机的嵌入式开发；
- 熟练使用 SW、UG 等三维建模软件，制图员四级，能够进行简单的结构设计与装配工作；
- 了解 Visual Studio 软件的使用与 C#的初步开发。

个人总结

- 基础扎实、求知欲强，富于团队合作精神，做事踏实认真，有强烈的责任心；
- FPGA 项目经验较丰富，具有较强的钻研能力，能够独立解决问题；具有团队合作精神与创新精神；
- 具备技术类文档的撰写能力，具有良好的沟通协作能力、较强的文字综合和表达能力；

Zhang Zhenhui

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Personal blog: **https://zhenhuizhang.tk**

Gender: Male | Native place: Fuzhou, Fujian

EDUCATION

Xidian University Sep 2019–Jul 2022

Space Engineering | M.E. | Space Science and Technology College Xi'an City

GPA: 3.60

Civil Aviation University of China Sep 2015–Aug 2019

Communication Engineering | B.E. | Electronic Information and Automation College Tianjin City

GPA: 3.46

HONORARY AWARDS

Best Idea /Best Hands on Ability Award of Honeywell-Star Jul 2017/Jul 2016

First Prize of College Students' Engineering Training Comprehensive Ability Competition May 2017

First/Third Prize of National Undergraduate Electronic Design Competition in Tianjin Dec 2017/Dec 2016

"Beidou Cup" Science and Technology Competition National Second/Excellent Prize May 2017/May 2016

Second Prize of NXP National College Students' Smart Car Competition in North China Aug 2017

First Prize of Tianjin Robot Competition Oct 2016

Second/Third Prize of Robot Competition in North China Nov 2017/ Nov 2016

PROJECT EXPERIENCE

Main control unit of scientific electronic control SPU Dec 2019-Up to Now

-- Shanghai Institute of Optics and precision machinery, Chinese Academy of Sciences

-- Head of Hardware Design

Project Introduction: the equipment will be used in the ultracold atomic physics experiment cabinet in the experimental module II "Mengtian" of China space station. By taking advantage of the microgravity environment, the equipment will create new extreme conditions, obtain the ultra-low temperature of 10-12K (PK) level that cannot be achieved on the ground, and build a super low temperature, large-scale and high-quality laboratory, An open experimental system of Bose and Fermi quantum degenerate working matter suitable for long time precise measurement. The controller is the main control module in the experimental system, which mainly realizes the functions of sending control commands to the laser subsystem and physical control unit, as well as the acquisition and processing of experimental data and images. Several boards are interconnected by VPX structure, and use several zynq ultrascale series FPGAs of Xilinx company as processing and control equipment. The configuration and control instructions required by the laser subsystem and physical control unit are transmitted through the network, and the experimental images are collected by usb-ccd camera, converted into HDMI signals and output to the display screen. Through the coordination of high-speed ADC, DAC and DDS devices, the laser intensity and frequency are monitored by AOM devices, and then the laser intensity and frequency are adjusted and stabilized by adjusting the output of DAC and DDS devices.

Main Tasks: as the project Hardware Director, complete the schematic drawing of all boards and PCB layout and wiring, including FPGA power supply circuit, PHY circuit, USB circuit; The hardware logic development of zynq device is completed. The user-defined IP based on Axi bus is written in Verilog language, and the axi-dma interconnection IP is added. The high-precision parallel work of multi-channel USB, network port, HDMI display, ADC, DAC, TTL, UART and SPI is realized. In addition, according to the special requirements of the space project, the circuit design and code writing are processed and verified.

Ultra high speed data acquisition and continuous storage system Jun 2019-May 2020

-- Xi'an Branch of China Academy of space technology (504 Institute)

-- FPGA Hardware Development Engineer

Project Description: the system uses two Xilinx virtex7 series FPGAs as the platform, uses two High-Speed ADCs with 5gsps sampling rate and 8bit data bit width for analog signal sampling, and transmits 80gbps sampling data to the computer memory for back-end processing in real time through PCIe.

Main work: as a member of the project team, completed the hardware logic development. Two vc709 development boards and four FMC sub boards of DSP 5gsps ADC are used for development and debugging. For a single FPGA development board, idelayer and iserders are used for high-speed data synchronization, and clock edge alignment algorithm is designed to eliminate metastable state; The multi-channel data automatic alignment algorithm is designed for the uneven data; The high-speed LVDS data received by FPGA is recombined with the idea of data splicing; The offset error, gain error and clock phase error of each ADC are eliminated to ensure equal interval sampling; Using MIG IP core to realize the control design of high-speed DDR3 chip, to meet the requirements of high-speed data storage and reading; The asynchronous problem of high speed parallel data transmission across clock domain is eliminated; With the cooperation of members in the same group, Xilinx's DMA for pcieip core is used for the development of high-speed data transmission of pciegen3 to realize 40Gbps stable data transmission and data dropping. Based on the principle of parallel alternating sampling, the clock with different phases is input to the two boards, and the data collected by the two boards are synchronized in the computer to realize the 80gbps data acquisition of the whole system.

A320-IPT Simulator Equipment Sep 2016-Sep 2019

-- Civil Aviation University of China

-- Project Leader

Project Description: During the University, participated in the research and development of QA320-IPT simulator equipment, as the project leader of the national project "building a simple aircraft simulator platform based on FSUIPC", and published a paper. So far, we have completed the design and processing of saddle frame, A320 throttle imitation, and combined with FSUIPC protocol to realize the docking with P3D flight simulation software.

PROFESSIONAL SKILLS

-- Proficient in using vivado software and Verilog hardware description language to develop FPGA devices, master the development process of FPGA and zynq devices of Xilinx company, and have rich experience in FPGA device design and development;

-- Proficient in the schematic design of various devices, able to independently carry out Multisim circuit simulation and ad circuit board design and drawing, with rich experience in circuit board design and development;

-- Familiar with the basic operation of Linux operating system and the development and design process of petalinux system, master the embedded development of 51 series, ardrino, STM32 Series MCU;

-- Proficient in using SW, UG and other three-dimensional modeling software, draftsman level 4, able to carry out simple structural design and assembly work;

-- Understand the use of Visual Studio software and the preliminary development of C #.

PERSONAL SUMMARY

-- Solid foundation, strong thirst for knowledge, team spirit, down-to-earth and serious work, strong sense of responsibility;

-- Have rich project experience about FPGA, strong research ability, and can solve problems independently; Team spirit and innovation spirit;

-- Have the ability to write technical documents, good communication and cooperation ability, strong text synthesis and expression ability;